

VisionInspect AI

Frontend Development Progress Report — Milestone 3

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1. Project Description

VisionInspect AI is an AI-powered quality inspection system that automates defect detection in manufactured products using computer vision and deep learning. Users upload product images and receive defect predictions, heatmaps, and quality decisions.

My role covers Frontend Development, API Integration, Authentication, and Role-Based Access Control (RBAC). For Milestone 3, the frontend has been fully connected to the backend and the AI prediction model, and the application is now completely functional end to end.

2. Technologies Used

- React.js (Vite)
- Tailwind CSS
- React Router
- React Context API
- JWT Authentication
- Flask REST APIs
- SQLite Database
- Git & GitHub

3. Work Completed

Authentication

User registration, login, JWT-based authentication, protected routes, and secure logout are fully implemented and working with the backend.

Role-Based Access Control

Two roles are supported — Quality Inspector (Dashboard, Inspection, History) and Factory Supervisor (Dashboard, Inspection, History, Analytics, Reports). Navigation and access dynamically adjust based on the logged-in role.

Dashboard

Displays total inspections, pass/fail counts, average quality score, and recent inspection activity, all fetched live from the backend.

Inspection

Fully integrated with the AI prediction system. Users upload an image and receive the defect category, confidence score, anomaly score, severity level, heatmap, and final quality decision.

History

Shows past inspection records with search, status filtering, and pagination. Quality Inspectors see their own records; Factory Supervisors see all records.

Analytics

Visualizes pass/fail statistics, severity distribution, and quality trends using live data from the backend.

Reports

Presents inspection summaries and quality statistics in a structured, downloadable layout.

4. API Integration

Method	Endpoint	Purpose
POST	/api/auth/login	User Authentication
POST	/api/auth/register	User Registration
POST	/api/inspection/predict	AI Inspection
GET	/api/history	Inspection History
GET	/api/analytics	Analytics
GET	/api/analytics/by-status	Status Analytics
GET	/api/reports	Reports

5. Testing Performed

- Registration, login, and invalid-login handling
- JWT authentication and role-based access
- Image upload and AI prediction with heatmap generation
- History retrieval, search, and filtering
- Analytics display and API error handling

6. Challenges Faced

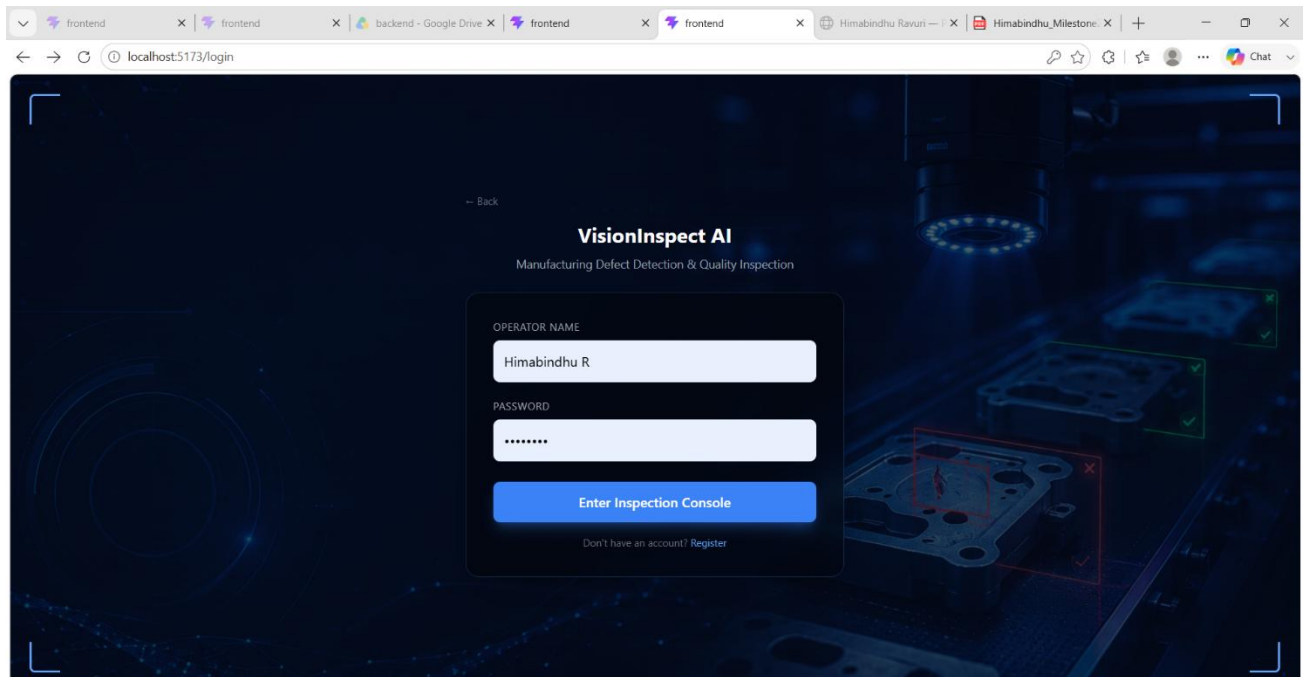
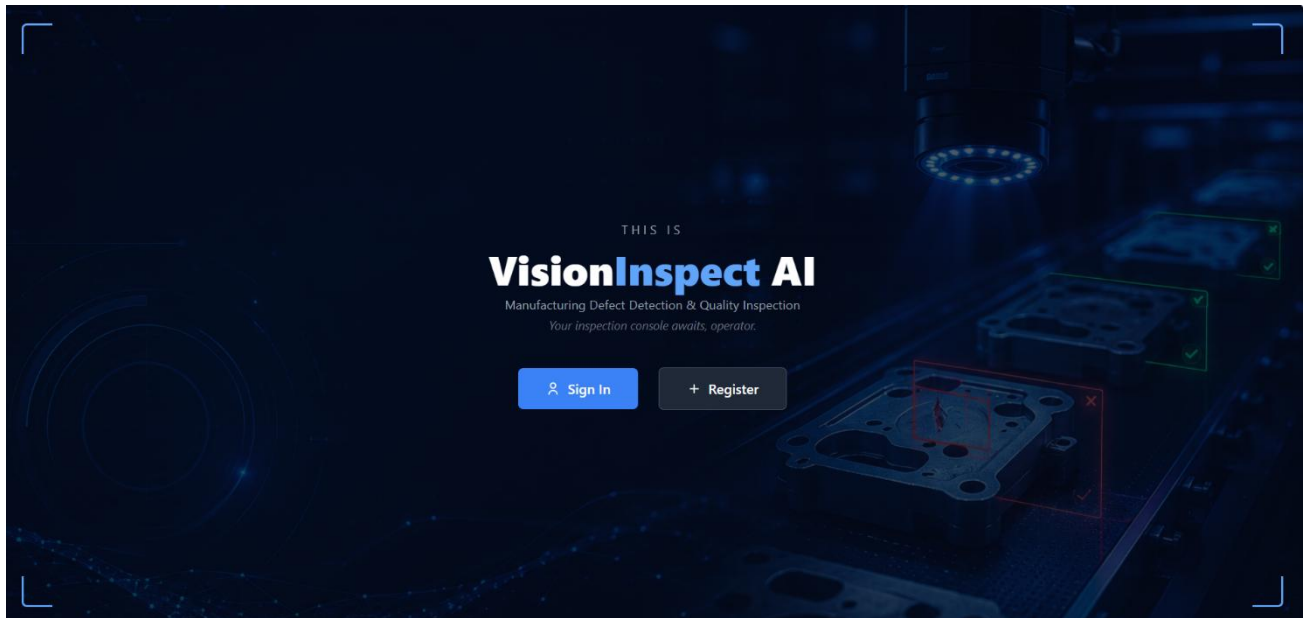
- Backend API integration and database synchronization issues
- Role-based access implementation
- AI prediction response and heatmap display handling
- Frontend-backend communication issues

All issues above were identified, debugged, and resolved through backend fixes, API testing, and frontend updates.

7. Screenshots

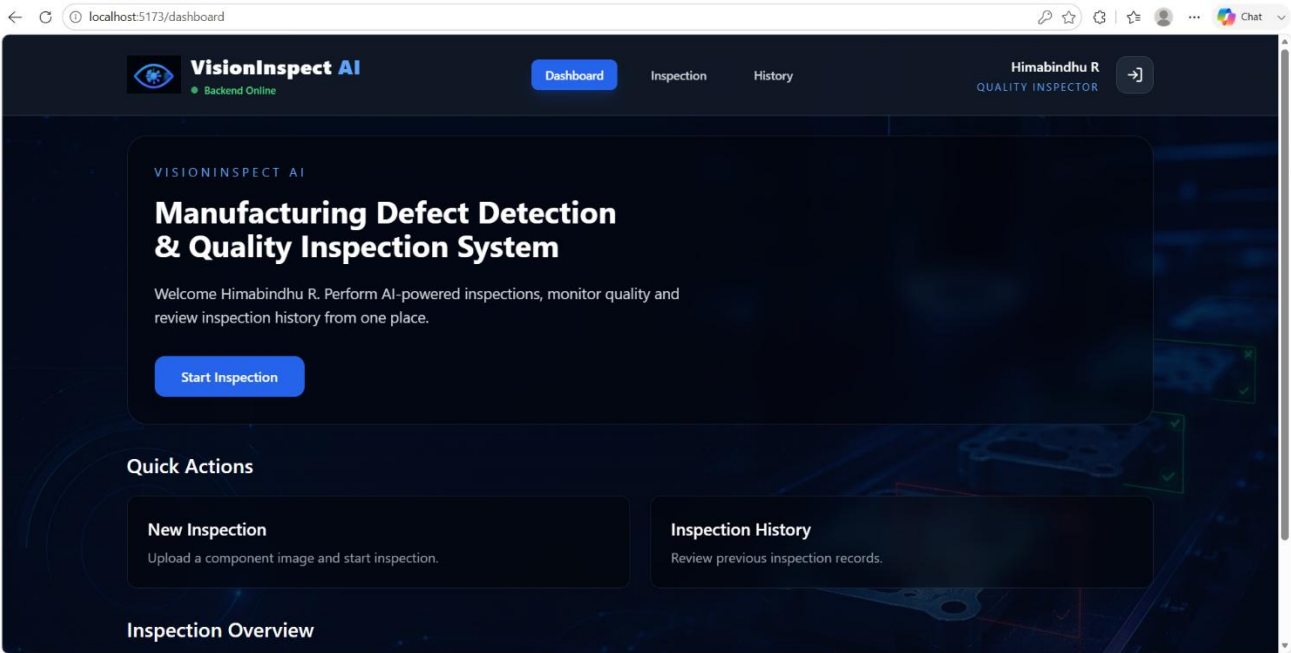
The following screenshots are included to document the completed application:

Login Page

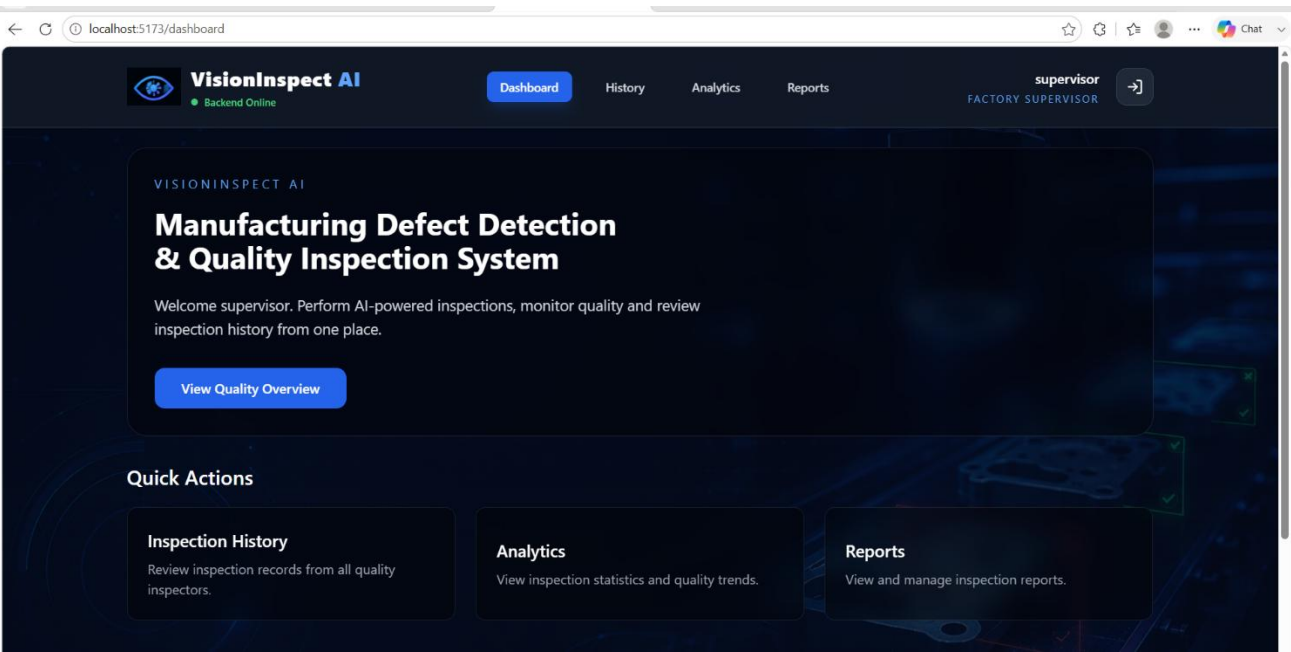


Role-Based Dashboards (Quality Inspector / Factory Supervisor)

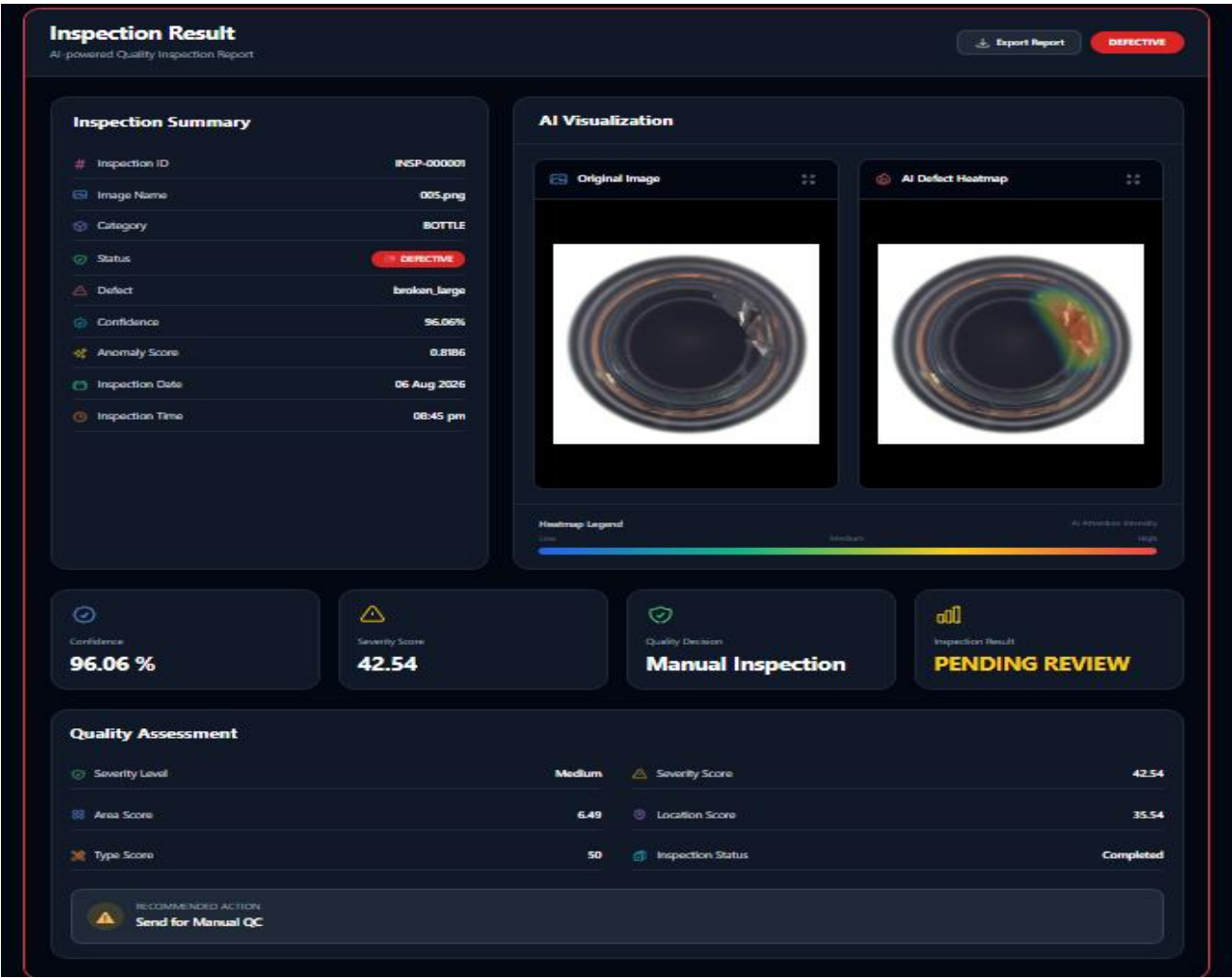
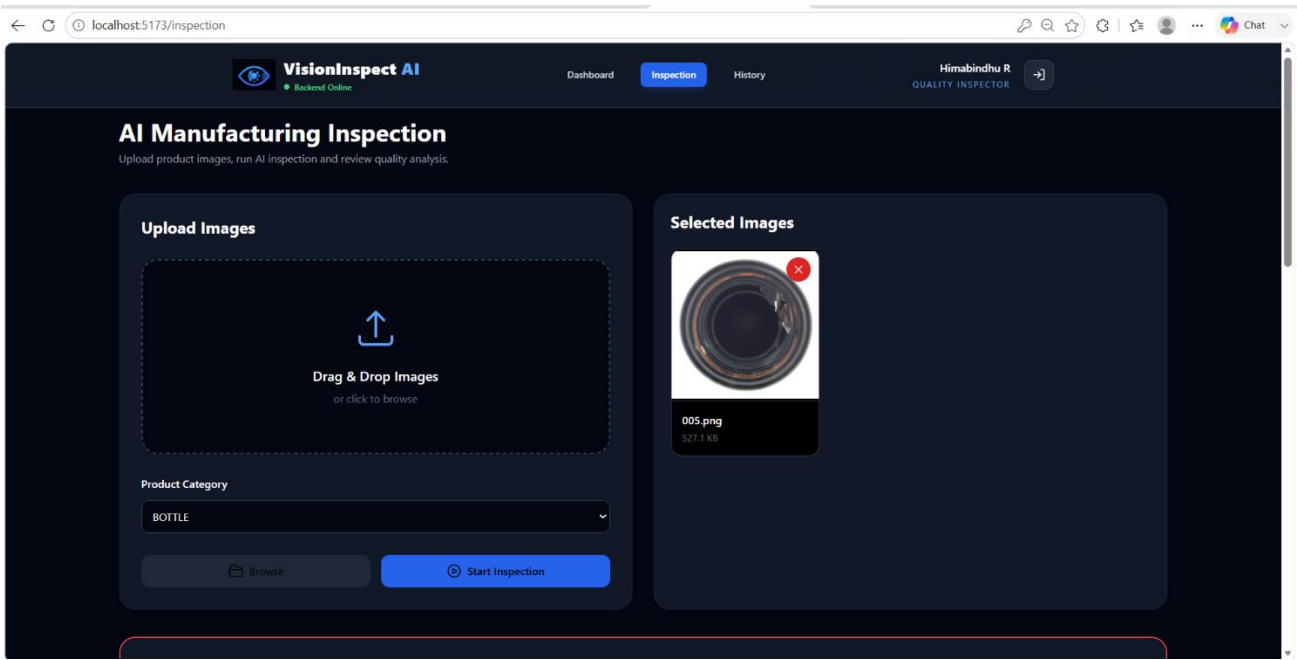
Quality Inspector



Factory Supervisor

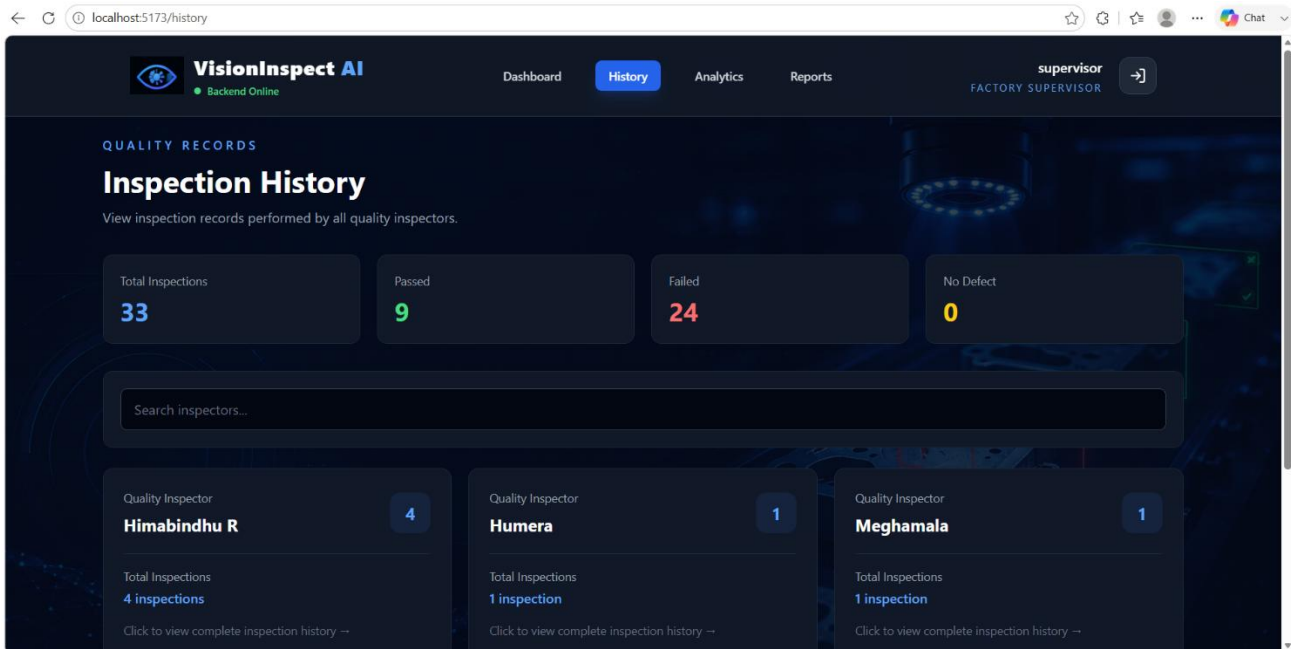


Inspection Upload & AI Prediction Result and Heatmap Output

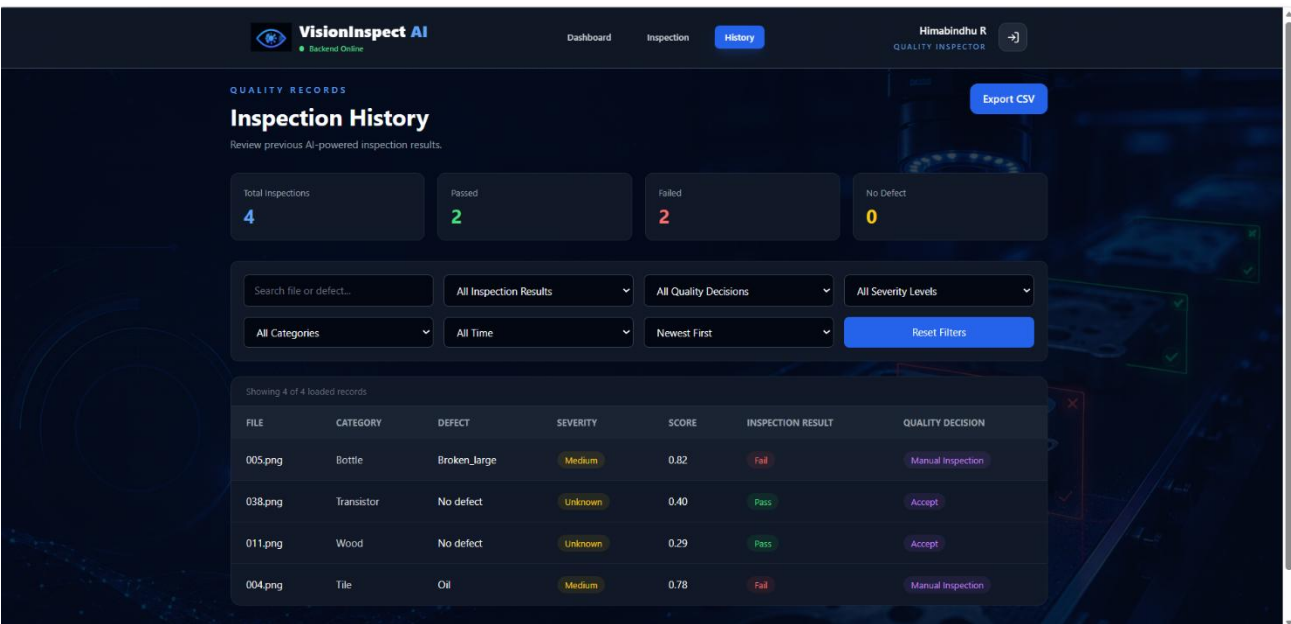


History Page with Search & Filter

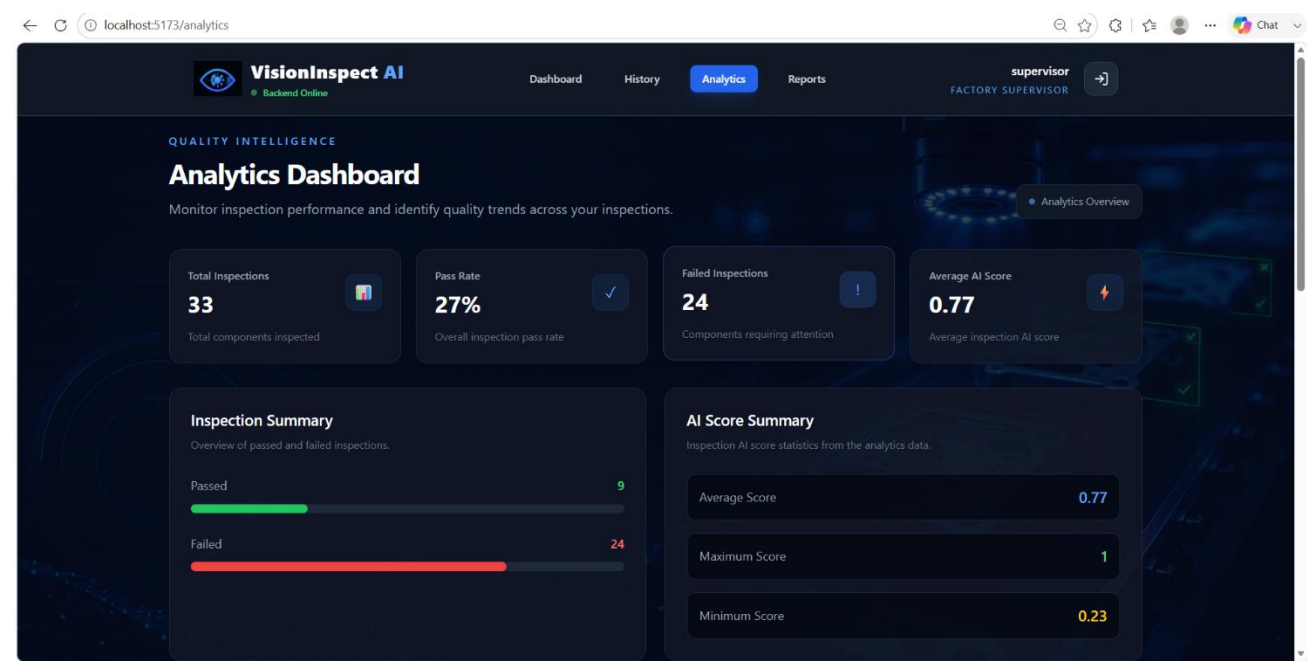
Supervisor History



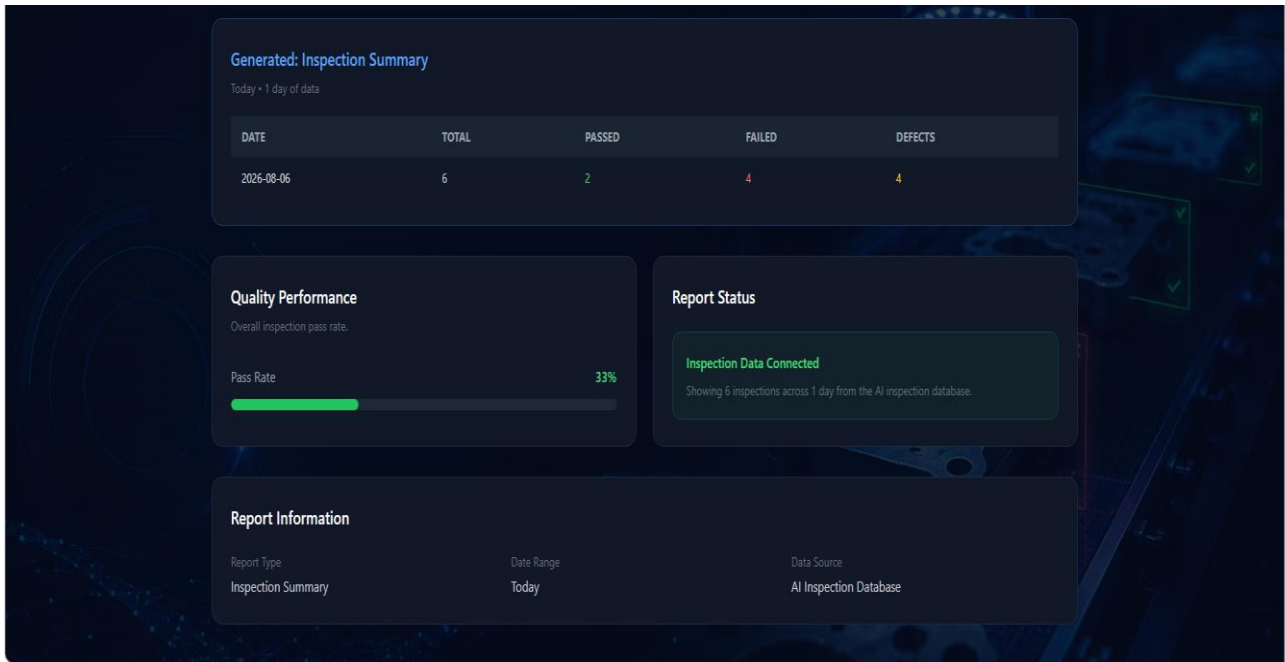
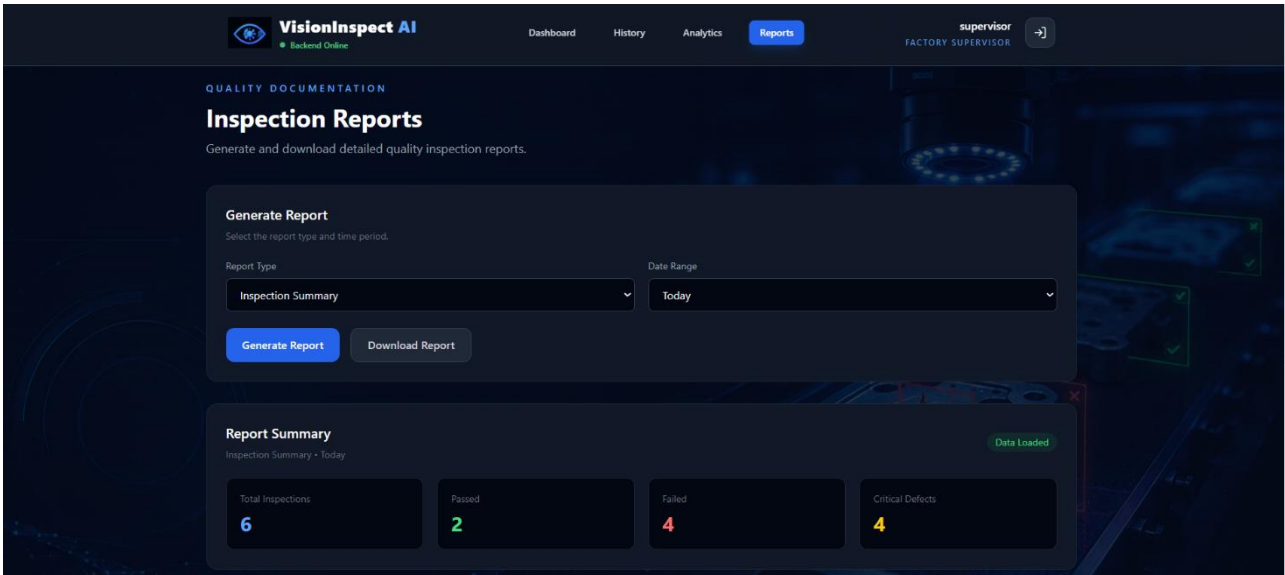
Inspector History



Analytics Dashboard



Reports Page



8. Final Outcome

VisionInspect AI is now fully functional, with the frontend, backend, and AI model completely integrated.

- Authentication & JWT Security
- Role-Based Access Control
- Dashboard
- Inspection with AI Prediction
- Heatmap Visualization
- History
- Analytics
- Reports
- Full API Integration & Error Handling

9. Conclusion

The VisionInspect AI frontend has been successfully completed and integrated with the backend and AI prediction model. The application now supports secure authentication, role-based access, real-time inspection, heatmap visualization, history management, analytics, and reporting — resulting in a complete, working, end-to-end quality inspection system ready for demonstration.